ACCESS TRADES

Trades Numeracy Answer Key

Answers to every Try it set and worksheet in the Trades Numeracy Textbook, and to every exercise in the Trades Numeracy Exercise Book, with working for the longer problems.

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How to use this key

The key follows the books in order. For each chapter, Part 1 gives the answers to the textbook's Try it sets and Shape Canvas worksheets, and Part 2 gives the answers to the Exercise Book sets A, B and C.

Answers that were rounded give the same number of decimal places the question asked for. Where a problem takes more than one step, the working is shown in smaller type under the answer. Your working can be different and still be right if it reaches the same answer.

1 Whole Numbers

ANSWERS

Part 1: Textbook

Try it 1.1

1. Three thousand forty-eight
2. 7,602
3. 5,000
4. 90
5. 2,450 mm (45 tens is more than 40 tens)
6. 20,050

Try it 1.2

1. 1,370
2. 1,380
3. 6,100
4. 6,000
5. 50,000
6. 800

Try it 1.4

1. 3,825
2. 1,950
3. 3,050 mm
4. 25,458

Try it 1.5

1. 3,125
2. 2,490
3. 1,125 mm
4. 1,200 mm

Try it 1.6

1. 120
2. 1,200
3. 8,000
4. 8,100 mm
5. 21 mm
6. 84

Try it 1.7

1. 12
2. 13
3. 125
4. 7 pieces (2,660 mm), offcut 340 mm
5. 4
6. 110 mm (550 ÷ 5)

Worksheet 1A

1. 2,050 mm
2. 1,075 mm
3. 4,410 mm
4. 15 pieces, offcut 300 mm
5. 991 mm (5,000 mm of pieces plus 9 mm of kerf)
6. 150 mm

Worksheet 1B

1. 600 mm
2. 700 mm
3. 900 + 250 + 600 + 450 + 300 + 700 = 3,200 mm
4. 3 × 8 = 24 holes, 24 × 2 = 48 minutes
5. Estimate 4 × 1,200 = 4,800. Exact 4,720.
6. 304 mm

Part 2: Exercise Book

Exercise 1A

1. **Four thousand seventy-two**
2. **15,308**
3. **7,000**
4. **3,049, 3,409, 3,490, 3,940**
5. **3,460**
6. **9,000**
7. **15,000**
8. **800**
9. **4,700 (1,200 + 2,900 + 600)**
10. **1,000 (50 × 20)**
11. **5,223**
12. **3,313**
13. **4,632**
14. **5,375**
15. **480**
16. **5,160**
17. **224**
18. **15 remainder 200**
19. **65**
20. **4,632 + 2,368 = 7,000**

Exercise 1B

1. **1,525 mm**
   Total cut 4,475 mm. 6,000 – 4,475 = 1,525.
2. **61 ft. 61 ÷ 21 = 2.9, so buy 3 lengths.**
   Round up: a part length still has to be bought.
3. **112 bolts, 5 boxes**
   14 × 8 = 112. 112 ÷ 25 = 4.48, round up to 5 boxes.
4. **144 nuts**
5. **13 pieces, offcut 6 in**
   240 ÷ 18 = 13.33. 13 × 18 = 234. 240 – 234 = 6.
6. **7 in**
   48 – 3 – 3 = 42 in. 7 holes make 6 spaces. 42 ÷ 6 = 7.
7. **8 hours**
8. **Estimate 50 × 24 = 1,200. Exact 1,080 parts.**
9. **5 boxes, 34 rods left over**
   18 × 12 = 216 rods. 216 ÷ 50 = 4.32, round up to 5 boxes = 250 rods. 250 – 216 = 34.

Exercise 1C

1. **1,625 mm**
2. **4,210 mm**
3. **1,365 mm**
4. **1,080 mm**
   6 spaces × 160 = 960. 960 + 60 + 60 = 1,080.
5. **750 mm**
6. **500 mm**
7. **1,100 + 300 + 750 + 500 + 350 + 800 = 3,800 mm**
8. **4 × 6 = 24 holes, 24 × 3 = 72 minutes**
9. **75 mm shorter. The offcut is 1,550 mm.**

2 Your Calculator: TI-30Xa and Sharp EL-531XT

ANSWERS

Part 1: Textbook

Try it 2.4

1. 81, √x (answer 9)
2. √, 81, = (answer 9)
3. 225
4. 14
5. 64
6. 4

Try it 2.5

1. 7
2. −24
3. −32
4. −4

Try it 2.6

1. 160
2. 81
3. 1,000
4. 628.32
5. 5,026.55
6. 3

Try it 2.7

1. $4\frac{1}{4}$ (4.25)
2. 0.625
3. 2.375
4. 90

Try it 2.10

1. 0.5
2. 0.7071
3. 0.3640
4. 30°
5. 36.87°
6. 191.51

Worksheet 2A

1. 2,845. TI-30Xa: 2450, +, 1375, −, 980, =. Sharp: 2450, +, 1375, −, 980, =.
2. 324. TI-30Xa: 18, x^2. Sharp: 18, x^2, =.
3. 23. TI-30Xa: 529, √x. Sharp: √, 529, =.
4. 12. TI-30Xa: 12, +⇄−, ×, 4, +, 60, =. Sharp: +/−, 12, ×, 4, +, 60, =.
5. D = 120 mm. π × 120 = 376.99 mm. TI-30Xa: π, ×, 120, =. Sharp: π, ×, 120, =.
6. $4\frac{3}{8}$ in. The TI-30Xa shows 4␣3␣8 and the Sharp shows 4 ┌3 ┌8. TI-30Xa: 2, a b/c, 3, a b/c, 4, +, 1, a b/c, 5, a b/c, 8, =. Sharp: 2, a b/c, 3, a b/c, 4, +, 1, a b/c, 5, a b/c, 8, =.
7. 68 kg. TI-30Xa: 340, ×, 20, 2nd, %, =. Sharp: 340, ×, 20, 2ndF, %.
8. 338.09. TI-30Xa: 25, SIN, ×, 800, =. Sharp: sin, 25, ×, 800, =.

Worksheet 2B

1. A: a b/c, the fraction key. B: +⇄−, change sign (after the digits). C: DRG, changes the angle unit. D: STO, stores the display in memory. E: 2ndF, for the orange second functions. F: +/−, change sign (before the number). G: √, square root (before the number). H: DEL, deletes a character.
2. $3\frac{5}{16}$
3. The square root of 225. The answer is 15.
4. Press DRG until DEG shows (RAD, then GRAD, then DEG: two presses).

Part 2: Exercise Book

Exercise 2A

1. **4,147. TI-30Xa: 3640, +, 1295, −, 788, =. Sharp: 3640, +, 1295, −, 788, =.**
2. **529. TI-30Xa: 23, x^2. Sharp: 23, x^2, =.**
3. **37. TI-30Xa: 1369, √x. Sharp: √, 1369, =.**
4. **343. TI-30Xa: 7, 2nd, x^3. Sharp: 7, x^3, =.**
5. **7. TI-30Xa: 343, 2nd, $\sqrt[3]{x}$. Sharp: 2ndF, $\sqrt[3]{}$, 343, =.**
6. **−50. TI-30Xa: 15, +⇄−, ×, 6, +, 40, =. Sharp: +/−, 15, ×, 6, +, 40, =.**
7. **243. TI-30Xa: 3, y^x, 5, =. Sharp: 3, y^x, 5, =.**
8. **282.74. TI-30Xa: π, ×, 90, =. Sharp: π, ×, 90, =.**
9. **$3\frac{13}{16}$ (TI-30Xa 3⌟13⌟16, Sharp 3 ┌13 ┌16). TI-30Xa: 2, a b/c, 3, a b/c, 8, +, 1, a b/c, 7, a b/c, 16, =. Sharp: 2, a b/c, 3, a b/c, 8, +, 1, a b/c, 7, a b/c, 16, =.**
10. **0.875. TI-30Xa: 7, a b/c, 8, 2nd, F⇄D. Sharp: 7, a b/c, 8, =, a b/c.**
11. **81. TI-30Xa: 450, ×, 18, 2nd, %, =. Sharp: 450, ×, 18, 2ndF, %.**
12. **1,350. TI-30Xa: 1250, +, 8, 2nd, %, =. Sharp: 1250, +, 8, 2ndF, %.**
13. **0.7880. TI-30Xa: 52, SIN. Sharp: sin, 52, =.**
14. **30.96°. TI-30Xa: 0.6, 2nd, TAN^{-1}. Sharp: 2ndF, $\tan^{-1}$, 0.6, =.**
15. **1,375. TI-30Xa: 640, +, 385, =, STO, 1, ON/C, 2400, −, RCL, 1, =. Sharp: 640, +, 385, STO, A, ON/C, 2400, −, RCL, A, =.**
16. **0.125. TI-30Xa: 8, 1/x. Sharp: 8, 2ndF, x^{-1}, =.**
17. **−16. Use the change-sign key, not the minus key. TI-30Xa: 25, +⇄−, +, 9, =. Sharp: +/−, 25, +, 9, =.**
18. **1,369 mm^2. TI-30Xa: 37, x^2. Sharp: 37, x^2, =.**

Exercise 2B

1. **359.1 mm**
   TI-30Xa: π, ×, 114.3, =. Sharp: π, ×, 114.3, =.
2. **1,130,973 mm^2**
   TI-30Xa: 600, x^2, ×, π, =. Sharp: 600, x^2, ×, π, =.
3. **$28\frac{1}{8}$ in**
   TI-30Xa: 72, −, 3, ×, 14, a b/c, 5, a b/c, 8, =. Sharp: 72, −, 3, ×, 14, a b/c, 5, a b/c, 8, =.
4. **425.6 m**
   TI-30Xa: 380, +, 12, 2nd, %, =. Sharp: 380, +, 12, 2ndF, %.
5. **75,600 mm**
   One frame: 4,200 mm. TI-30Xa: 1240, +, 860, +, 1240, +, 860, =, STO, 1, ON/C, RCL, 1, ×, 18, =. Sharp: 1240, +, 860, +, 1240, +, 860, STO, A, ON/C, RCL, A, ×, 18, =.
6. **1,500 mm**
   TI-30Xa: (, 1200, x^2, +, 900, x^2,), √x. Sharp: √, (, 1200, x^2, +, 900, x^2,), =.
7. **10.6°**
   TI-30Xa: (, 450, ÷, 2400,), 2nd, TAN^{-1}. Sharp: 2ndF, $\tan^{-1}$, (, 450, ÷, 2400,), =.
8. **−15 (15 below the setting)**
   TI-30Xa: 11, +⇄−, −, 4, =. Sharp: +/−, 11, −, 4, =.

Exercise 2C

1. **A: π. B: x^2, squares a number. C: RCL, recalls a memory. D: 2nd, for the second functions printed above the keys. E: y^x, powers. F: a b/c, the fraction key. G: RCL, recalls a memory. H: HOME, returns to normal mode.**
2. **$6\frac{11}{16}$**
3. **18,240**
4. **$3^4 = 81$**
5. **20,000,000,000 (2×10^{10})**
6. **One press (the order is DEG, RAD, GRAD, then back to DEG).**
7. **D = 100 mm. π × 100 = 314.16 mm. Doubled: π × 200 = 628.32 mm, twice as far.**

3 Order of Operations

ANSWERS

Part 1: Textbook

Try it 3.2

1. 18
2. 17
3. 18
4. 20

Try it 3.3

1. 10
2. 5
3. 45
4. 4,000

Try it 3.4

1. 16
2. 18
3. 36
4. 18

Try it 3.5

1. 24
2. 6
3. 32
4. 25

Try it 3.6

1. 9
2. 306
3. 20
4. 400

Worksheet 3A

1. 135,000 + 225,000 = 360,000 mm²
2. 2 × (600 + 250) = 1,700 mm. With L = 1,200 mm: 2 × (1,200 + 250) = 2,900 mm.
3. π × (1,600 − 625) = π × 975 = 3,063.05 mm²
4. (1,350 − 150) ÷ 8 = 150 mm

Worksheet 3B

1. 16
2. 16
3. 18 and 36. In 2 × 3² the square belongs to the 3 only.
4. 32 (16 ÷ 4 = 4, 4 × 2 = 8, 40 − 8 = 32)
5. (12 + 8) ÷ 4 = 5
6. 3 × (5 − 2) + 4 = 13
7. 10 and 14
8. (45 + 27) ÷ (14 − 6) = gives 9
9. 6,000 − (4,500 + 9) = 1,491 mm
10. 5 × 4² = 80. It grew by 35 (from 45 to 80).
11. The calculator divided 60 ÷ 3 first. Key (120 + 60) ÷ 3 = to get 60 m per job.

Part 2: Exercise Book

Exercise 3A

1. 33
2. 44
3. 26
4. 32
5. 4
6. 3
7. 36
8. 16
9. 20
10. 48
11. 10
12. 16
13. 4
14. 25
15. 30
16. 29
17. 12
18. 2.5
19. (4 + 6) × 3 = 30
20. 24 ÷ (6 − 2) = 6

Exercise 3B

1. 6 × 8 + 4 = 52 bolts
2. 4,450 mm
   Brackets first: 2,225. Then 2 × 2,225.
3. 6 in
   4 × 27 = 108, 2 × 15 = 30, 108 + 30 = 138, 144 − 138 = 6.
4. 3,111.67 mm
5. 13 in
6. 2,840 mm
7. Right: 3 × 8 + 2 × 10 = 44 hours. The wrong one gives 260.
8. The calculator divided 300 ÷ 4 first. Key (1500 − 300) ÷ 4 = to get 300 mm.

Exercise 3C

1. 140,000 + 300,000 = 440,000 mm²
2. L = 1,500 − 600 = 900 mm
3. 1,188 mm
4. π × (3,600 − 2,025) = 4,948.01 mm². With r = 30 mm: π × (3,600 − 900) = 8,482.30 mm².
5. 230 mm

4 Fractions

ANSWERS

Part 1: Textbook

Try it 4.1

1. $\frac{3}{4}$
2. 7
3. 16
4. $\frac{5}{12}$ used, $\frac{7}{12}$ left
5. $\frac{1}{8}$ in
6. No. The numerator 9 is bigger than the denominator 8, so it is worth more than one whole.

Try it 4.2

1. $\frac{7}{16}$ in
2. $1\frac{1}{4}$ in
3. $1\frac{15}{16}$ in
4. $2\frac{5}{8}$ in
5. $3\frac{3}{16}$ in
6. $3\frac{7}{8}$ in

Try it 4.3

1. $\frac{5}{8}$
2. $\frac{7}{8}$
3. $\frac{5}{8}$
4. $\frac{3}{4}$
5. $\frac{10}{16}$
6. $\frac{24}{32}$
7. $\frac{7}{16}$
8. $\frac{5}{32}$
9. $\frac{5}{8}$, $\frac{11}{16}$, $\frac{3}{4}$
10. $\frac{2}{3}$ ($\frac{16}{24}$ is bigger than $\frac{15}{24}$)

Try it 4.4

1. $1\frac{3}{4}$
2. $1\frac{13}{16}$
3. $2\frac{5}{8}$
4. $\frac{19}{8}$
5. $\frac{11}{2}$
6. $\frac{31}{16}$

Try it 4.5

1. 2 a b/c 7 a b/c 16
2. $1\frac{3}{8}$
3. $\frac{15}{16}$
4. $\frac{7}{8}$
5. $\frac{19}{4}$
6. $\frac{5}{8}$

Try it 4.6

1. $\frac{13}{16}$
2. $1\frac{3}{8}$
3. $17\frac{5}{8}$
4. $52\frac{13}{16}$ in
5. $\frac{9}{16}$
6. $3\frac{5}{8}$
7. $6\frac{9}{16}$
8. $13\frac{5}{16}$ in

Try it 4.7

1. $1\frac{1}{2}$
2. 12
3. $3\frac{3}{8}$
4. $14\frac{1}{4}$ in
5. $\frac{8}{5}$
6. $\frac{7}{16}$
7. 6
8. $9\frac{3}{4}$ in

Worksheet 4A

1. A: $\frac{13}{16}$ in, B: $1\frac{3}{8}$ in, C: $2\frac{15}{16}$ in
2. D: $\frac{5}{32}$ in, E: $\frac{11}{32}$ in
3. F: $\frac{11}{64}$ in, G: $\frac{29}{64}$ in
4. $1\frac{3}{8}$ in is 6 sixteenths past the 1 in line. $2\frac{11}{16}$ in is 11 sixteenths past the 2 in line.
5. $21\frac{3}{8}$ in

   $96 - (24\frac{3}{8} + 31\frac{1}{2} + 18\frac{3}{4}) = 96 - 74\frac{5}{8} = 21\frac{3}{8}$
6. $11\frac{15}{16}$ in

   $40 - 12\frac{5}{8} - 15\frac{7}{16} = 11\frac{15}{16}$
7. $21\frac{1}{2}$ in

   $144 - 2 \times 36\frac{3}{8} - 2 \times 24\frac{5}{8} - 4 \times \frac{1}{8} = 21\frac{1}{2}$
8. 43 in in total. $\frac{1}{2}$ in more, for a new total of $43\frac{1}{2}$ in.
9. $14\frac{3}{8}$ in

   $72 \div 5 = 14\frac{2}{5}$. $\frac{2}{5} \times 16 = 6.4$, nearest 6 sixteenths.

Worksheet 4B

1. $7\frac{1}{4}$ in

   $10\frac{1}{2} - 3\frac{1}{4} = 7\frac{1}{4}$
2. $8\frac{3}{4}$ in

   $6\frac{1}{8} + 2\frac{5}{8} = 8\frac{3}{4}$
3. $1\frac{3}{16}$ in

   $\frac{6}{16} + \frac{8}{16} + \frac{5}{16} = \frac{19}{16} = 1\frac{3}{16}$
4. 12 pieces, offcut $1\frac{1}{2}$ in

   Each piece uses $2\frac{1}{4} + \frac{1}{8} = 2\frac{3}{8}$ in. $30 \div 2\frac{3}{8} = 12\frac{12}{19}$, so 12 pieces. Offcut $30 - 12 \times 2\frac{3}{8} = 1\frac{1}{2}$
5. $2\frac{3}{8}$ in

   $10\frac{1}{2} - 3\frac{3}{8} - 4\frac{3}{4} = 2\frac{3}{8}$
6. $12\frac{1}{4}$ in
7. $5\frac{1}{4}$ in

   $(36 - 2 \times 2\frac{1}{4}) \div 6 = \frac{63}{2} \div 6 = 5\frac{1}{4}$

Part 2: Exercise Book

Exercise 4A

1. $\frac{5}{12}$. **Yes, the numerator is smaller than the denominator.**
2. $\frac{19}{8} = 2\frac{3}{8}$
3. **A:** $\frac{9}{16}$ **in, B:** $1\frac{5}{8}$ **in, C:** $2\frac{1}{4}$ **in, D:** $2\frac{13}{16}$ **in**
4. **E:** $\frac{7}{32}$ **in, F:** $\frac{15}{32}$ **in**
5. **G:** $\frac{37}{64}$ **in, H:** $\frac{53}{64}$ **in**

   G is 5 small spaces past the $\frac{1}{2}$ in line ($\frac{32}{64}$). H is 5 small spaces past the $\frac{3}{4}$ in line ($\frac{48}{64}$).
6. $\frac{7}{16}$ **and** $\frac{9}{16}$
7. $\frac{28}{32}$ **and** $\frac{20}{64}$
8. $\frac{5}{8}$ **in, by** $\frac{1}{16}$ **in**
9. $\frac{3}{8}$, $\frac{13}{32}$, $\frac{7}{16}$
10. $4\frac{5}{8}$ **and** $3\frac{5}{16}$
11. $\frac{71}{16}$ **and** $\frac{27}{4}$
12. $3\frac{9}{16}$. **The TI-30Xa shows 57␣16, which is** $\frac{57}{16}$.
13. $\frac{2}{5}$

    $2\frac{1}{2} = \frac{5}{2}$, turned upside down is $\frac{2}{5}$
14. **(a)** $1\frac{1}{16}$ **(b)** $10\frac{5}{16}$
15. **(a)** $3\frac{5}{16}$ **(b)** $4\frac{3}{8}$
16. **(a)** $7\frac{1}{2}$ **(b)** $4\frac{3}{8}$
17. **(a) 4 (b)** $5\frac{1}{4}$

Exercise 4B

1. $15\frac{5}{16}$ **in**

   $96 - (27\frac{5}{8} + 19\frac{3}{16} + 33\frac{1}{2}) - 3 \times \frac{1}{8} = 96 - 80\frac{5}{16} - \frac{3}{8}$

   $= 15\frac{5}{16}$
2. $41\frac{3}{16}$ **in**
3. $1\frac{1}{2}$ **in**
4. **6 shims**

   $\frac{3}{16} \div \frac{1}{32} = \frac{3}{16} \times \frac{32}{1} = 6$
5. $7\frac{13}{16}$ **in**

   $15\frac{5}{8} \div 2 = \frac{125}{8} \times \frac{1}{2} = \frac{125}{16}$
6. $5\frac{9}{16}$ **in**

   $(48 - 2 \times 1\frac{3}{4}) \div 8 = 44\frac{1}{2} \div 8 = 5\frac{9}{16}$
7. **15 spacers, offcut** $2\frac{1}{2}$ **in**

   Each spacer uses $4\frac{3}{8} + \frac{1}{8} = 4\frac{1}{2}$ in. $70 \div 4\frac{1}{2} = 15\frac{5}{9}$, so 15 spacers. $15 \times 4\frac{1}{2} = 67\frac{1}{2}$ in, offcut $2\frac{1}{2}$ in.
8. **2 in**

   $2\frac{3}{8} - 2 \times \frac{3}{16} = 2\frac{3}{8} - \frac{3}{8}$
9. **No. The key is** $\frac{1}{16}$ **in too wide.**
10. **1,800 mm**
11. **30 in and 15 in**
12. $16\frac{11}{16}$ **in**

    $50 \div 3 = 16\frac{2}{3}$. $\frac{2}{3} \times 16 = 10.67$, nearest 11 sixteenths.

Exercise 4C

1. **$8\frac{7}{8}$ in**

$12\frac{1}{4} - 3\frac{3}{8}$

2. **$6\frac{5}{8}$ in**

$9\frac{1}{2} - 2\frac{7}{8}$

3. **$4\frac{1}{4}$ in**

$14\frac{1}{2} - 2\frac{5}{8} - 3\frac{1}{4} - 4\frac{3}{8} = 4\frac{1}{4}$

4. **$1\frac{7}{8}$ in**

$(30 - 5 \times 5\frac{1}{4}) \div 2$

5. **$33\frac{1}{2}$ in, then $33\frac{3}{4}$ in**

$4 \times 8\frac{3}{8} = 33\frac{1}{2}$. $4 \times \frac{1}{16} = \frac{1}{4}$ more.

6. **$1\frac{1}{16}$ in**

$\frac{4}{16} + \frac{10}{16} + \frac{3}{16} = \frac{17}{16}$

7. **Along: $7\frac{1}{2}$ in. Across: $2\frac{1}{4}$ in.**

$9\frac{3}{4} - 2 \times 1\frac{1}{8}$ and $4\frac{1}{2} - 2 \times 1\frac{1}{8}$

5 Decimals

ANSWERS

Part 1: Textbook

Try it 5.1

1. Three and six hundredths
2. 5 thousandths (0.005)
3. 3
4. 0.9
5. 0.45
6. $\frac{125}{1{,}000}$

Try it 5.2

1. 0.6
2. 0.125, 0.25, 0.5
3. 6.4
4. 9.53
5. 3.0 mm
6. 19 mm

Try it 5.3

1. 5.25
2. 22.225
3. 6.75
4. 57.15
5. 219 mm
6. 0.85 mm

Try it 5.4

1. 0.25
2. 7.125
3. 0.18
4. 475
5. 609.6 mm
6. 15.875 mm

Try it 5.5

1. 2.4
2. 5
3. 50.8
4. 6.35
5. 152.4 mm
6. 33.33 mm

Try it 5.6

1. 0.75
2. 1.875 in
3. $\frac{5}{8}$
4. $3\frac{3}{8}$ in (0.4 × 16 = 6.4, rounds to 6)
5. $\frac{19}{32}$ in (0.59 × 32 = 18.88, rounds to 19)
6. $3\frac{15}{16}$ in (100 ÷ 25.4 × 16 = 62.99, rounds to 63)

Try it 5.7

1. $14.74
2. $18.75
3. $2.16
4. $13.18
5. $12.35
6. $13.68

Worksheet 5A

1. 88.9 mm
   165.1 − 31.75 − 44.45 = 88.9
2. A = 13 mm, B = 28.5 mm, C = 36.5 mm
3. 1,057.2 mm
   2,438.4 − (457.2 + 609.6 + 304.8) − 3 × 3.2 = 1,057.2
4. 52.5 mm
   60.3 − 2 × 3.9 = 52.5
5. 457.2 mm
   152.4 + 76.2 + 152.4 + 76.2
6. 3.05 mm. It is more than 3 mm.
7. 186.27 mm
   (1,219.2 − 2 × 50.8) ÷ 6 = 1,117.6 ÷ 6 = 186.27

Worksheet 5B

1. **4.375 in = $4\frac{3}{8}$ in, 2.625 in = $2\frac{5}{8}$ in**
2. **2.875 in = $2\frac{7}{8}$ in**

   6.5 – 1.25 – 2.375 = 2.875
3. **$1\frac{11}{16}$ in (0.71 × 16 = 11.36, rounds to 11) and $1\frac{23}{32}$ in (0.71 × 32 = 22.72, rounds to 23)**
4. **$2\frac{15}{16}$ in**

   75 ÷ 25.4 × 16 = 47.24, rounds to 47
5. **7.94 mm**

   5 ÷ 16 × 25.4 = 7.9375
6. **$79.26**

   3 × 6 × 4.15 + 12 × 0.38 = 74.70 + 4.56
7. **14.25 in = $14\frac{1}{4}$ in. New total 13.5 in = $13\frac{1}{2}$ in.**

Part 2: Exercise Book

Exercise 5A

1. **4 hundredths (0.04). Seventy-five thousandths.**
2. **0.03 and 0.048**
3. **A = 8.5 mm, B = 26 mm, C = 44.5 mm, D = 57 mm**
4. **0.375, 0.4, 0.41**
5. **2.5 mm, by 0.05 mm**
6. **7.8 and 7.85**
7. **31.5 mm and 19 mm**

   31.27 × 2 = 62.54, rounds to 63, ÷ 2 = 31.5. 18.8 × 2 = 37.6, rounds to 38, ÷ 2 = 19
8. **73.83**
9. **32.54**
10. **(a) 0.24 (b) 177.8 (c) 3,125**
11. **(a) 70 (b) 7.62 (c) 0.195**
12. **0.875 and 3.25**
13. **$\frac{7}{8}$ and $\frac{2}{25}$**

    $\frac{875}{1{,}000} \div 125 = \frac{7}{8}$. $\frac{8}{100} \div 4 = \frac{2}{25}$
14. **$1\frac{1}{2}$ in (0.47 × 16 = 7.52, rounds to 8) and $1\frac{15}{32}$ in (0.47 × 32 = 15.04, rounds to 15)**
15. **$\frac{5}{16}$ in (0.29 × 16 = 4.64, rounds to 5)**
16. **$1\frac{3}{16}$ in**

    30 ÷ 25.4 × 16 = 18.90, rounds to 19. $\frac{19}{16} = 1\frac{3}{16}$
17. **31.75 mm**

    1.25 × 25.4

Exercise 5B

1. **142.75 mm**

   1,500 – (412.5 + 287.25 + 650) – 3 × 2.5 = 1,500 – 1,349.75 – 7.5
2. **$81.03**

   3 × 1.85 = 5.55 m. 5.55 × 14.60
3. **7 shims**

   1.75 ÷ 0.25 = 175 ÷ 25
4. **$14.50**
5. **10.62 kg**
6. **117.86 mm**

   (900 – 2 × 37.5) ÷ 7 = 825 ÷ 7
7. **$3\frac{3}{8}$ in**

   0.38 × 16 = 6.08, rounds to 6. $\frac{6}{16} = \frac{3}{8}$
8. **$4\frac{3}{4}$ in**

   120 ÷ 25.4 × 16 = 75.59, rounds to 76. $\frac{76}{16} = 4\frac{3}{4}$
9. **$802.50**
10. **15.875 mm**

    0.625 × 25.4

Exercise 5C

1. **102.25 mm**
   210 − 45.5 − 62.25
2. **88.9 mm**
   77.9 + 2 × 5.5
3. **101.6 mm**
   (609.6 − 2 × 203.2) ÷ 2
4. **45 mm**
   (750 − 4 × 165) ÷ 2
5. **1,143 mm, then 1,135.5 mm**
   5 × 228.6. 5 × 1.5 = 7.5 mm less.
6. **5.875 in = $5\frac{7}{8}$ in, 3.25 in = $3\frac{1}{4}$ in**
7. **A = 23.5 mm, B = 41 mm, 17.5 mm apart**

6 Percent, Ratio and Proportion

ANSWERS

Part 1: Textbook

Try it 6.1

1. 0.42
2. 0.075
3. 6%
4. 125%
5. $\frac{2}{5}$
6. 62.5%

Try it 6.2

1. 90
2. 100
3. 48
4. 75
5. 20 bolts
6. 609.6 mm

Try it 6.3

1. 25%
2. 3%
3. 15%
4. 120
5. 300
6. 25%

Try it 6.4

1. 69 m
2. 27 sheets (26.4 rounded up)
3. $200.00
4. 1,260 mm
5. 15%
6. 20%

Try it 6.5

1. 2 : 3
2. 3 : 1
3. 1 : 3
4. 600 mL and 300 mL
5. 1,500 mL and 500 mL
6. 5 : 3

Try it 6.6

1. 150 mm per minute
2. 12
3. 24 minutes
4. B: $6.90 per m (A is $7.20 per m)
5. $130.50
6. 3

Try it 6.7

1. Yes. 3 × 20 = 4 × 15 = 60
2. x = 18
3. x = 16
4. 21 m
5. 10 hours
6. 0.6 L

Try it 6.8

1. 480 mm
2. 1,800 mm
3. 60 mm
4. 60 mm
5. 75 : 1,500 = 1 : 20
6. 11 ft

Worksheet 6A

1. 35%, 0.35, $\frac{7}{20}$
2. 500 mm
3. 138 m

   120 × 1.15
4. 25% decrease
5. 900 mm, then 720 mm. It is not back to 750 mm.

Worksheet 6B

1. **1,800 mL of base and 600 mL of hardener**
   2,400 mL ÷ 4 parts = 600 mL per part
2. **960 mm by 600 mm**
3. **90 mm**
   40/30 = 120/x, so x = 30 × 120 ÷ 40
4. **96 mm. Every line is twice as long as on the 1 : 20 drawing.**
5. **13 ft**
   3.25 ÷ 0.25
6. **125 L per minute, 18 minutes**

Part 2: Exercise Book

Exercise 6A

1. **0.64**
2. **0.035**
3. **70%**
4. **240%**
5. $\frac{9}{20}$
6. **87.5%**
7. **78**
8. **273**
9. **7.5%**
10. **120**
11. **391**
12. **1,150**
13. **25%**
14. **3 : 7**
15. **1 : 6**
    250 : 1,500, divide both by 250
16. **750 and 450**
    1,200 ÷ 8 = 150 per part
17. **70 L per minute**
18. **x = 12**
19. **3,400 mm**
20. **42 mm**

Exercise 6B

1. **71.68 m, 12 lengths**
   64 × 1.12 = 71.68, ÷ 6.4 = 11.2, round up
2. **21 sheets (20.7 rounded up)**
3. **$76.12 each, $456.72 for 6**
   86.5 – 12% = 76.12, × 6
4. **0.4 m per s, 125 s**
5. **7.06%**
6. **24.5 m**
   9.8 × 35 ÷ 14
7. **1 L and 0.4 L**
   1.4 ÷ 7 = 0.2 L per part
8. **43 ft**
   5.375 ÷ 0.125
9. **15%**
10. **5 : 2**
    300 : 120, divide both by 60

Exercise 6C

1. **62%, 0.62,** $\frac{31}{50}$
2. **560 mm**
3. **2,640 mm, 55%**
4. **60 mm**
   50/20 = 150/x
5. **1,400 mm by 800 mm**
6. **0.6 L of A and 0.9 L of B**
7. **3.75 ft = 3′ 9″**
8. **80 mm × 50 mm at 1 : 10, 160 mm × 100 mm at 1 : 5. Every line is twice as long.**

7 Measurement and Unit Conversion

ANSWERS

Part 1: Textbook

Try it 7.1

1. 3,600 mm
2. 0.85 m
3. 420 mm
4. 6.096 m
5. 2,500 mL
6. 1.8 t

Try it 7.2

1. 67 in
2. $122\frac{1}{4}$ in
3. 7′ 2″
4. $10' 5\frac{1}{2}''$
5. 12 ft
6. $8' 4\frac{1}{4}''$

Try it 7.3

1. 304.8 mm
2. 7.9375 mm
3. 6.096 m
4. 30 in
5. $78\frac{3}{4}''$
6. 49.90 kg

Try it 7.4

1. 3,000,000 mm²
2. 0.45 m²
3. 720 in²
4. 6,451.6 mm²
5. 480 L
6. 3,456 in³

Try it 7.5

1. 37.85 L
2. 45.46 L
3. 95.10 US gal
4. 1.201 US gal
5. 7.48 US gal
6. 109.98 imp gal

Try it 7.6

1. 68 °F
2. 14 °F
3. 100 °C
4. 10 °C
5. 204.4 °C
6. 27 °F

Worksheet 7A

1. 1,219.2 mm by 762 mm
2. 24″ by 14″
   610 ÷ 25.4 = 24.0157, 355 ÷ 25.4 = 13.9764
3. $11' 9\frac{1}{4}''$ = 3,587.75 mm
4. Length 1,981.2 mm, width 63.5 mm, thickness 9.525 mm
5. 68.04 kg
6. 2.4 m, 240 cm, $7' 10\frac{1}{2}''$
   2,400 ÷ 25.4 = 94.4882 in

Worksheet 7B

1. 200 L
2. 18.70 US gal
   30 × 12 × 12 = 4,320 in³, ÷ 231
3. 1,080,000 mm² = 1.08 m²
   2 × (600 × 400 + 600 × 300 + 400 × 300)
4. 302 °F. A drop of 72 °F.
5. 1 L, then 8 L (8 times as much)

Part 2: Exercise Book

Exercise 7A

1. **4,750 mm**
2. **0.625 m**
3. **380 mm**
4. **3.25 L**
5. **2,400 kg**
6. **0.75 kg**
7. **101 in**
8. **115 3/4 in**
   9 × 12 + 7.75 = 115.75
9. **7′ 10″**
10. **13′ $1\frac{1}{2}$″**
11. **15.875 mm**
12. **57.15 mm**
13. **$37\frac{3}{8}$″**
    950 ÷ 25.4 = 37.4016, × 16 = 598.43, about 598 sixteenths
14. **4.572 m**
15. **34.02 kg**
16. **18 ft, 5.4864 m**
17. **2,500,000 mm²**
18. **864 in²**
19. **16,129 mm²**
20. **96 L**
21. **75.71 L**
22. **113.65 L**
23. **95 °F**
24. **−20 °C**

Exercise 7B

1. **7′ 4″ (88″)**
   2,235 ÷ 25.4 = 87.9921 in
2. **14′ $9\frac{1}{4}$″ = 4,502.15 mm**
   54.5 + 87.75 + 35 = 177.25 in
3. **119.75 kg**
4. **27.9 m² = 300.31 ft², buy 1 US gal**
   27.9 ÷ 0.092 903 04, then ÷ 350 = 0.858
5. **243 L = 64.19 US gal**
6. **36.03 US gal**
   30 × 4.546 09 ÷ 3.785 411 784
7. **93.3 °C and 248 °F**
8. **63 °F**
9. **90 in³ = 1,474.84 cm³**
10. **2.9768 m² = 32.04 ft²**

Exercise 7C

1. **914.4 mm by 609.6 mm**
2. **Hole 20.6375 mm, width 101.6 mm, thickness 6.35 mm**
3. **14′ $0\frac{1}{2}$″ = 4,279.9 mm**
   50.5 + 81.25 + 36.75 = 168.5 in
4. **300 mm**
   150 L = 150,000,000 mm³, ÷ (1,000 × 500)
5. **432 in² = 3 ft²**
6. **9 ft² = 1,296 in²**
7. **70 °C = 158 °F, and −4 °F = −20 °C**
8. **0.125 L, then 1 L, 8 times as much**

8 Formulas and Transposing

ANSWERS

Part 1: Textbook

Try it 8.1

1. V
2. π (the 2 in r^2 is a power, it tells you to square r)
3. C and D
4. 2 × l × w
5. s × s
6. 1.8 and 32

Try it 8.2

1. 2.88 m^2
2. 2,100 mm
3. 942.48 mm
4. 1,256.64 cm^2
5. 1.5 m^3
6. 95 °F

Try it 8.3

1. 82
2. 200
3. 250
4. 140
5. 4,725
6. 250

Try it 8.4

1. l = A ÷ w
2. 2.4 m
3. 48 in (4 ft)
4. 300 mm
5. 84 mm
6. 0.5 m
7. 600 mm

Try it 8.5

1. 700 mm
2. w = P ÷ 2 – l
3. 100 °C
4. –40 °C
5. 8 hours, from h = (T – 60) ÷ 85
6. 100 mm

Try it 8.6

1. 40 cm
2. 0.6 m
3. 100 mm
4. 50.46 mm
5. 1.91 m, from h = V ÷ (π × r^2)
6. The side becomes 3 times as long (√9 = 3).

Worksheet 8A

1. 60 cm
   w = A ÷ l = 7,200 ÷ 120
2. 45 cm
   s = √A = √2,025
3. r = 39.89 mm, D = 79.79 mm
   r = √(5,000 ÷ π), D = 2 × r
4. 240 mm
   h = 2 × A ÷ b = 2 × 43,200 ÷ 360
5. 0.75 m
   h = V ÷ (l × w) = 0.72 ÷ (1.2 × 0.8)
6. 1.77 m
   h = V ÷ (π × r^2) = 2 ÷ (π × 0.6^2)
7. 90 cm. The side doubled.
8. 30 cm. The width was halved.
9. l × w must be in brackets: h = V ÷ (l × w) = 0.75 m.
   Check: 1.2 × 0.8 × 0.75 = 0.72.

Worksheet 8B

1. D = C ÷ π = 636.62 mm
2. w = P ÷ 2 – l = 900 mm
3. b = 2 × A ÷ h = 450 mm
4. $t_C = (t_F - 32) \div 1.8 = 232.22$ °C
5. h = (T – 60) ÷ 85 = 5.5 hours

Part 2: Exercise Book

Exercise 8A

1. **Subject V. Variables V, r and h. Constant π.**
2. **3 × a × b and π × r × r**
3. **9 m²**
4. **4,250 mm**
5. **1,413.72 mm**
6. **3,848.45 mm²**
7. **0.81 m³**
8. **14 °F**
   1.8 × (−10) = −18, −18 + 32 = 14
9. **465**
10. **150**
11. **250**
12. **270**
13. **90**
14. **12.25**
15. **l = A ÷ w**
16. **D = C ÷ π**
17. **w = V ÷ (l × h)**
18. **s = P ÷ 4**
19. **l = P ÷ 2 − w**
20. **$t_C = (t_F - 32) \div 1.8$**
21. **b = 2 × A ÷ h**
22. **h = (T − a) ÷ r**
23. **s = √A**
24. **r = √(A ÷ π)**
25. **h = V ÷ (π × r²)**
26. **n = L ÷ (π × D)**

Exercise 8B

1. **0.9 m**
   l = A ÷ w = 0.54 ÷ 0.6
2. **1.2 m**
   h = V ÷ (l × w) = 1.8 ÷ (1.5 × 1.0)
3. **123.61 mm**
   r = √(12,000 ÷ π) = 61.80, D = 2 × r
4. **1,950 mm**
   l = P ÷ 2 − w = 3,200 − 1,250
5. **121.11 °C**
   (250 − 32) ÷ 1.8
6. **36 in = 3 ft**
   s = √1,296
7. **2.26 m**
   h = 4 ÷ (π × 0.75²)
8. **13 days**
   d = (285.50 − 45) ÷ 18.50
9. **$16\frac{11}{16}$″**
   D = 52.375 ÷ π = 16.671 in, 0.671 × 16 = 10.74, round to 11
10. **200 mm**
    b = 2 × A ÷ h
11. **0.282 m**
    r = √(0.25 ÷ π)
12. **The part is 1.04 m², less than the 1.08 m² shelf, but it is 0.65 m deep and the shelf is only 0.6 m deep, so it does not fit.**
13. **96 in = 8 ft**
    l = A ÷ w = 4,608 ÷ 48
14. **31.83 turns**
    150 ft = 1,800 in, n = 1,800 ÷ (π × 18)
15. **Learner: 4,000 − 1,400 = 2,600 mm. Check: 2 × (2,600 + 700) = 6,600, not 4,000. Correct: l = P ÷ 2 − w = 1,300 mm.**

Exercise 8C

1. **60 cm**
2. **56 mm**
3. **r = 50.46 mm, D = 100.93 mm**
4. **180 mm**
5. **1.2 m**
6. **0.44 m**
   $r = \sqrt{(V \div (\pi \times h))} = \sqrt{(0.9 \div (\pi \times 1.5))}$
7. **450**
8. **477.46 mm**
9. **168 mm, three times the side**
10. **201.85 mm. The diameter doubled.**

9 Perimeter and Circumference

ANSWERS

Part 1: Textbook

Try it 9.1

1. 3,400 mm
2. 3.6 m
3. 750 mm
4. 640 mm
5. 144 in = 12 ft
6. 500 mm
7. $121\frac{1}{2}$ in

Try it 9.2

1. 90 mm
2. 84.15 mm
3. 12 in
4. The circumference, a little more than 3 times longer (π times).

Try it 9.3

1. 628.32 mm
2. 376.99 mm
3. $31\frac{7}{16}$"
4. 795.77 mm, from D = C ÷ π
5. 400 mm
6. 15,707.96 mm
7. 2,000 mm

Try it 9.4

1. 785.40 mm
2. 1,286.64 mm
3. $14\frac{1}{8}$"
4. 300 mm

Try it 9.5

1. 785.40 mm
2. 314.16 mm
3. 471.24 mm
4. 235.62 mm

Try it 9.6

1. 162.83 mm
2. 1,914.16 mm (four quarter circles make one circle, D = 100 mm)
3. 1,671.24 mm
4. 1,071.24 mm

Worksheet 9A

1. 5,500 mm
2. 42 cm
3. 750 mm
   w = P ÷ 2 – l = 4,200 ÷ 2 – 1,350
4. 2 × (124 + 80) = 408 in = 34 ft
5. 500 mm
   s = P ÷ 4
6. 11,000 mm. The perimeter doubled.

Worksheet 9B

1. C = π × 350 = 1,099.56 mm
2. D = 3,000 ÷ π = 954.93 mm
3. π × 400 = 1,256.64 mm
4. 75 ÷ 360 × π × 800 = 523.60 mm
5. 4.55 turns
6. $27\frac{1}{8}$"

Worksheet 9C

1. 2 × 80 + π × 40 = 285.66 mm
2. 500 + 500 + 240 + π × 240 ÷ 2 = 1,616.99 mm
3. 2 × 720 + 2 × 420 + π × 80 = 2,531.33 mm
4. π × 120 + π × 60 = 565.49 mm
5. 2 × 1,000 + π × 250 = 2,785.40 mm
6. 2 × 250 + π × 500 ÷ 4 = 500 + 392.70 = 892.70 mm

Part 2: Exercise Book

Exercise 9A

1. **7,100 mm**
2. **300 cm**
3. **3.2 m**
4. **450 mm**
5. **360 in = 30 ft**
6. **$50\frac{3}{4}$ in**
   2 × (15 6/8 + 9 5/8) = 2 × 25 3/8
7. **2 × (111 + 70) = 362 in = 30′ 2″**
8. **700 mm**
9. **136.5 mm**
10. **165 mm**
11. **1,193.81 mm**
12. **596.90 mm**
13. **477.46 mm**
14. **44″**
15. **314.16 mm**
16. **1,570.80 mm**
17. **1,005.31 mm**
18. **It is cut in half too.**

Exercise 9B

1. **2 × (30 + 21) = 102 in = 8′ 6″**
2. **5,962 mm**
   π × 1,850 + 150 = 5,961.95
3. **1,140 mm**
   π × 355 + 25 = 1,140.27
4. **$33\frac{3}{4}$″**
   π × 10.75 = 33.772 in, 0.772 × 16 = 12.35
5. **1,138.83 m**
   1,450 × π × 250 ÷ 1,000
6. **15.92 turns**
   100 ft = 1,200 in, 1,200 ÷ (π × 24)
7. **528.73 mm**
8. **650 mm**
   2,042 ÷ π = 649.99
9. **16,000 mm, so 3 lengths (2 lengths hold only 12,000 mm)**
10. **2,356.19 mm**
    90 ÷ 360 × π × 3,000
11. **Before 3,142.48 mm, after 3,442.48 mm, 300 mm longer (each straight run grows by 150 mm).**
12. **3,872.74 mm**
    π × 1,220 + 40
13. **2,050 mm**
    2 × (600 + 400) + 50

Exercise 9C

1. **1,900 mm**
2. **51 cm**
3. **π × 500 = 1,570.80 mm**
4. **120 ÷ 360 × π × 700 = 733.04 mm**
5. **2 × 600 + π × 250 = 1,985.40 mm**
6. **2 × 780 + 2 × 280 + π × 120 = 2,496.99 mm**
   Straight edges 900 − 2 × 60 = 780 and 400 − 2 × 60 = 280. Four quarter circles make one circle, D = 120
7. **885.84 mm**
   c = (L − π × D) ÷ 2
8. **π × 140 + π × 80 = 691.15 mm (was 659.73 mm)**

10 Area

ANSWERS

Part 1: Textbook

Try it 10.1

1. 28 cm²
2. Thirty-five square metres
3. 12 ft²
4. 30 ft²

Try it 10.2

1. 300,000 mm²
2. 62,500 mm²
3. 15 ft² = 2,160 in²
4. 7 tins (40 ÷ 6 = 6.67 L)

Try it 10.3

1. 30,000 mm²
2. 94.5 in²
3. 0.54 m²
4. 30,000 mm²

Try it 10.4

1. 150,000 mm²
2. 360 in²
3. 80,000 mm²
4. 162 in²

Try it 10.5

1. 20,106.19 mm²
2. 62,831.85 mm²
3. 876.50 mm²
4. 863.94 mm² (R = 30, r = 25)

Try it 10.6

1. 130,000 mm²
2. 99,893.81 mm²
3. 58,743.36 mm²
4. 250,000 mm²

Try it 10.7

1. 760,000 mm²
2. 470,000 mm²
3. 16.00 ft² (2,304 in²)
4. 314,159.27 mm²

Worksheet 10A

1. 750,000 mm² = 0.75 m²
2. 720 in² = 5 ft²
3. 280,000 mm²
4. 300 mm
   54,000 × 2 ÷ 360 = 300
5. 159,043 mm²
6. 14,137.17 mm²
   π × (90² – 60²) = π × 4,500

Worksheet 10B

1. B is 450 mm high. Area =
   90,000 + 90,000 = 180,000 mm²
2. Hidden width 600 mm. Area =
   75,000 + 75,000 + 90,000 = 240,000 mm²
3. 312 in² = 2.17 ft²
   240 + 72 = 312 in², 312 ÷ 144
4. 131,415.93 mm²
   A = 500 × 200, B + C = one full circle, π × 100²

Worksheet 10C

1. 1,840,000 mm² = 1.84 m²
2. 31,415.93 mm² and 125,663.71 mm². The new area is 4 times bigger (2 × 2).
3. Stretch-out π × 400 = 1,256.64 mm. SA =
   1,005,309.65 mm²
   Ends π × 200² × 2, side π × 400 × 600
4. 52 in across. 2,496 in² = 17.33 ft²

Part 2: Exercise Book

Exercise 10A

1. **45 cm²**
2. **1.5 m²**
3. **80,000 mm²**
4. **864 in², 3 ft²**
5. **294,000 mm²**
6. **2.56 m²**
7. **70,200 mm²**
8. **52.5 in²**
9. **0.3 m²**
10. **59,400 mm²**
11. **6,361.73 mm²**
12. **706.86 in²**
    r = 15 in
13. **2.26 m²**
14. **196,350 mm²**
15. **6,283.19 mm²**
    R = 60, r = 40, π × (3,600 − 1,600)
16. **The radii must be squared before subtracting: π × (60² − 40²), not π × (60 − 40)².**
17. **9 in**
    54 × 2 ÷ 12
18. **25,000 cm²**

Exercise 10B

1. **25.92 m², 9 sheets**
   7.2 × 3.6 = 25.92 m², one sheet = 2.88 m², 25.92 ÷ 2.88 = 9
2. **4.57 L, so 5 tins**
   Area 2 × 4 × 4 = 32 m², 32 ÷ 7
3. **52,778.76 mm²**
4. **2,304 in² = 16 ft²**
   16 × 12 ÷ 2 = 96 in² each
5. **200,979.47 mm²**
6. **3.24 m²**
   One side 810,000 mm²
7. **40 ft² (5,760 in²)**
8. **9.80 m²**
   Ends π × 0.6² × 2, side π × 1.2 × 2
9. **504,657.08 mm²**
10. **1.58 m²**
    Stretch-out π × 168, times 3,000
11. **900,000 mm² = 0.9 m²**
12. **133 ft²**
    9 ft 6 in = 9.5 ft, 14 × 9.5

Exercise 10C

1. **700 mm**
   420,000 ÷ 600
2. **500 mm**
   300,000 ÷ ((400 + 800) ÷ 2) = 300,000 ÷ 600
3. **100 mm**
   r = √(31,416 ÷ π)
4. **Foot 700 mm long. Area =**
   225,000 + 175,000 = 400,000 mm²
5. **180,000 + 30,000 = 210,000 mm²**
6. **40,796.46 mm²**
7. **1,370,000 mm² = 1.37 m²**
   2 × (360,000 + 225,000 + 100,000)
8. **Stretch-out 942.48 mm. SA = 612,610.57 mm²**
9. **90,000 mm² and 810,000 mm², 9 times bigger (3 × 3)**

11 Volume and Capacity

ANSWERS

Part 1: Textbook

Try it 11.1

1. 30 cm^3
2. 1,000 mm^3
3. 3,456 in^3
4. 8 m^3

Try it 11.2

1. 0.12 m^3
2. 5,184 in^3 = 3 ft^3
3. 0.3 m^3
4. 1,000 cm^3

Try it 11.3

1. 2,500 L
2. 200 L
3. 189.27 L
4. 52.83 US gal
5. 4 US gal (924 ÷ 231)
6. 28.32 L

Try it 11.4

1. 254.47 L
2. 78.54 L
3. 70.50 US gal
4. 17.67 L

Try it 11.5

1. 37.70 L
2. 48 L ($600^2 \times 400 \div 3 = 48{,}000{,}000\ mm^3$)
3. 480 in^3
4. 400 mm, the height at right angles to the base

Try it 11.6

1. 4.19 L
2. 56.55 L
3. 0.524 m^3
4. 904.78 in^3

Try it 11.7

1. 1,727.88 cm^3
2. 75.40 L
3. 252 L
4. 80 L (4 times)

Worksheet 11A

1. 0.2 m^3 = 200 L
2. 27,000,000 mm^3 = 27 L
3. 120 L
   $600 \times 400 \div 2 \times 1{,}000$
4. 600 mm
   480 L = 480,000,000 mm^3, $\div (1{,}000 \times 800)$
5. 157.08 L
6. 42.41 L

Worksheet 11B

1. 376.99 L
   $A = \pi \times 400^2 \times 600$, $B = \pi \times 400^2 \times 450 \div 3$
2. 261.80 L
   Cylinder $\pi \times 250^2 \times 1{,}000$ plus one sphere $4 \div 3 \times \pi \times 250^3$
3. 33.51 L
4. 9,600 $in^3 \div 231$ = 41.56 US gal

Worksheet 11C

1. 15.71 L and 62.83 L, 4 times bigger
2. 8 L and 64 L, 8 times bigger
3. 39.27 L and 157.08 L, 4 times bigger
   $\pi \times 0.025^2 \times 20 \times 1{,}000$ and $\pi \times 0.05^2 \times 20 \times 1{,}000$
4. 28.27 L
   The circles are 300 mm across, so r = 150 mm. The height is 400 mm

Part 2: Exercise Book

Exercise 11A

1. **72 cm³**
2. **3,000 mm³**
3. **2.5 m³**
4. **450 L**
5. **2 ft³**
   3,456 ÷ 1,728
6. **2 L**
7. **0.07 m³**
8. **0.216 m³**
9. **45 L**
10. **5 US gal**
11. **283.91 L**
12. **132.09 US gal**
13. **42.41 L**
14. **18.85 L**
15. **162 L**
16. **65.45 L**
17. **0.134 m³**
18. **5.03 L**
19. **0.6 m**
20. **7.48 US gal**
21. **The 40 L drum. 10 US gal = 37.85 L**
22. **0.063 m³**

Exercise 11B

1. **180 L, 9 pails**
2. **52.88 US gal**
   r = 9 in, π × 81 × 48 in³, ÷ 231
3. **159.28 L**
4. **363.25 L**
5. **900 L, 20 minutes**
6. **1,432.57 L**
   Cylinder 1.1310 m³ + cone 0.3016 m³
7. **57.66 US gal**
8. **2,094.40 L**
9. **1,099.56 L**
10. **5,890.49 cm³**
11. **12,214.51 cm³**
12. **74.81 US gal, about 30 hours**
13. **432 L**

Exercise 11C

1. **800 mm**
   240,000,000 mm³ ÷ (600 × 500)
2. **1,000 mm**
   h = V ÷ (π × r²)
3. **64 L**
   400³ = 64,000,000 mm³
4. **117.81 L**
   The circles are 500 mm across, so r = 250 mm
5. **179.07 L**
6. **134.04 L**
7. **End area 1,056 mm², volume 422.4 cm³**
   80 × 8 + 52 × 8 = 640 + 416
8. **4.19 L and 33.51 L, 8 times bigger (2 × 2 × 2)**
9. **28.27 L and 56.55 L, 2 times bigger**

12 Angles and Right Triangles

ANSWERS

Part 1: Textbook

Try it 12.1

1. 180°
2. 45°
3. 45°
4. 72°
5. 30°
6. 3/4 of a turn (270 ÷ 360)

Try it 12.2

1. acute, right, obtuse, acute, straight
2. 108°
3. 45°
4. 105°
5. 90°
6. obtuse

Try it 12.3

1. 60°
2. 63°
3. 45°
4. No. Two right angles already add to 180°, leaving 0° for the third angle.
5. 65° each
6. 60°

Try it 12.4

1. 1,000 mm
2. 1,749.29 mm
3. 1,200 mm
4. 1.73 m
5. 15″
6. $28\frac{5}{16}$″ (28.302″)

Try it 12.5

1. 500 mm
2. 15 ft
3. Closed (less than 90°). It should read 1,500 mm.
4. 2 m
5. 25″
6. Yes. $25 + 144 = 169 = 13^2$

Try it 12.6

1. 2,000 mm
2. 1,950 mm
3. 3,354 mm
4. 424.26 mm
5. No. The diagonals are not equal.
6. $87\frac{3}{4}$″

Worksheet 12A

1. 58°
2. 63°
3. 42°
4. 115°
5. 45°
6. 72° each
7. Two triangles hold 2 × 180° = 360°. ? = 360 − 85 − 100 − 75 = 100°
8. 125°, an obtuse angle
9. 36° with 10 holes, 30° with 12 holes. More holes give a smaller angle.

Worksheet 12B

1. 750 mm
2. 0.5 m
3. $26\frac{5}{8}$″ (26.627″)
4. 2,012.46 mm
5. 1,750 mm
6. It should read 1,500 mm. At 1,507 mm the corner is open past 90°.
7. 500 mm. With legs of 600 mm and 800 mm it is 1,000 mm, so it doubles too.
8. Bottom diagonal $\sqrt{(600^2 + 400^2)} = 721.11$ mm. Rod $\sqrt{(520{,}000 + 300^2)} = 781.02$ mm

Part 2: Exercise Book

Exercise 12A

1. **120°**
2. **40°**
3. **acute, right, obtuse, obtuse**
4. **116°**
5. **75°**
6. **57°**
7. **52°**
8. **58° each**
9. **110°**
10. **2,500 mm**
11. **594.05 mm**
    $\sqrt{(350^2 + 480^2)} = \sqrt{352{,}900}$
12. **1,500 mm**
    $\sqrt{(1{,}700^2 - 800^2)} = \sqrt{2{,}250{,}000}$
13. **2.19 m**
    $\sqrt{(2.6^2 - 1.4^2)} = \sqrt{4.8}$
14. **$23\frac{5}{8}$″ (23.601″)**
    $\sqrt{(196 + 361)} = \sqrt{557}$
15. **Yes. $64 + 225 = 289 = 17^2$**
16. **No. $36 + 81 = 117$, but $11^2 = 121$**
17. **1,253 mm (1,253.00 mm)**
18. **636.40 mm**
    $450 \times \sqrt{2}$

Exercise 12B

1. **30° between neighbours. Hole 1 to hole 4 is 3 spaces: 90°**
2. **38°, acute**
3. **750 mm each. Two braces and two cuts: $2 \times 750 + 2 \times 3 = 1{,}506$ mm**
   $\sqrt{(450^2 + 600^2)} = \sqrt{562{,}500} = 750$
4. **It should read 2 m. At 1.998 m it is closed, less than 90°.**
5. **3,499 mm each (3,498.57 mm). Push the two corners at the ends of the 3,505 mm diagonal.**
   $\sqrt{(3{,}000^2 + 1{,}800^2)} = \sqrt{12{,}240{,}000}$
6. **2,387.99 mm**
   $\sqrt{(2{,}250^2 + 800^2)} = \sqrt{5{,}702{,}500}$
7. **$91\frac{9}{16}$″, which is 7′ $7\frac{9}{16}$″**
   48″ and 78″: $\sqrt{(2{,}304 + 6{,}084)} = \sqrt{8{,}388} = 91.586$″
8. **$18 + 24 + 30 = 72$″**
   Hypotenuse $\sqrt{(18^2 + 24^2)} = 30$″
9. **120° evenly spaced. The two arms that did not move keep 120° between them, and the third angle is $360 - 100 - 120 = 140°$. The other two angles are 120° and 140°.**

Exercise 12C

1. **390 mm**
2. **1.5 m**
3. **62°**
4. **120°**
5. **57°**
6. **1,700 mm**
7. **721.11 mm. No, it did not double.**
8. **Legs 900 mm and $600 - 300 = 300$ mm. Sloped edge $\sqrt{(900^2 + 300^2)} = 948.68$ mm**
9. **Face diagonal 707.11 mm. Rod $\sqrt{(707.11^2 + 500^2)} = 866.03$ mm**

13 Trigonometry

ANSWERS

Part 1: Textbook

Try it 13.1

1. q
2. p
3. r
4. p
5. q
6. 59°

Try it 13.2

1. 0.6428
2. 0.7660
3. 0.8391
4. 1
5. 0.8660 and 0.8660. The same, because 60° and 30° are the two acute angles of one triangle.
6. sin A = 5 ÷ 13 = 0.3846, tan A = 5 ÷ 12 = 0.4167

Try it 13.3

1. 338.09 mm
2. 1,149.07 mm
3. 649.84 mm
4. 1,046.07 mm
5. 703.16 mm
6. $19\frac{5}{16}$″ (19.284″)

Try it 13.4

1. 32.01°
2. 44.43°
3. 38.21°
4. 36.87° and 53.13°
5. 30°
6. 45°

Try it 13.5

1. 4.29°, 7.50%
2. 2.86°
3. 150 mm
4. 209.27 mm
5. 26.57°
6. 11,449 mm

Try it 13.6

1. 848.53 mm
2. Travel 461.88 mm, run 230.94 mm
3. 653.28 mm
4. $14\frac{1}{8}$″
5. 450 mm
6. 45°

Try it 13.7

1. Corner 108°, cut 54°
2. 12 (each corner turns 30°, and 360 ÷ 30 = 12)
3. 11.55 mm
4. 29.36°
5. 70°
6. 92° (2 × 46°)

Worksheet 13A

1. 321.39 mm
2. 638.05 mm
3. 1.69 m
4. 1,307.59 mm
5. $12\frac{5}{16}$″ (12.313″)
6. $31\frac{3}{8}$″ (31.382″)

Worksheet 13B

1. 33.69°
2. 36.87°
3. 30.07°
4. 16.26° and 73.74°
5. Angle 4.76°, grade 8.33%
6. Angle 22.62°, sloped edge 13″

Worksheet 13C

1. **Travel $25\frac{7}{16}$″ (25.456″), run 18″**
2. **Travel 783.94 mm, run 724.26 mm**
3. **29.99°, so 30° fittings**
4. **True offset √(300² + 400²) = 500 mm, travel 707.11 mm**
5. **12.28 mm**
6. **Corner 135°, cut 67.5°**
7. **At 6°: 313.59 mm. At 12°: 623.74 mm. No, it is a little less than double (2 × 313.59 = 627.18).**

Part 2: Exercise Book

Exercise 13A

1. **Opposite M: e. Adjacent to M: d. Hypotenuse: f. Opposite N: d.**
2. **0.4226, 0.9063, 0.4663**
3. **sin A = 8/17 = 0.4706, cos A = 15/17 = 0.8824, tan A = 8/15 = 0.5333**
4. **344.45 mm**
   650 × sin 32°
5. **836.41 mm**
   1,250 × cos 48°
6. **400.71 mm**
   900 × tan 24°
7. **844.62 mm**
   520 ÷ sin 38°
8. **166.29 mm**
   300 ÷ tan 61°
9. **$10\frac{7}{8}$″ (10.870″)**
   42 × sin 15°
10. **23.63°**
    $\tan^{-1}(350 \div 800)$
11. **22.02°**
    $\sin^{-1}(0.45 \div 1.2)$
12. **42.54°**
    $\cos^{-1}(700 \div 950)$
13. **62.6°**
14. **Grade 2%, angle 1.15°**
15. **14.04°**

Exercise 13B

1. **Travel 530.33 mm, run 375 mm**
   375 ÷ sin 45°, and tan 45° = 1
2. **Travel $10\frac{3}{8}$″ (10.392″), run $5\frac{3}{16}$″ (5.196″)**
   9 ÷ sin 60°, 9 ÷ tan 60°
3. **$31\frac{3}{8}$″ (31.358″)**
   12 ÷ sin 22.5°
4. **Set 320.00 mm, run 554.26 mm**
   640 × sin 30°, 640 × cos 30°
5. **30.00°**
   $\tan^{-1}(250 \div 433)$
6. **True offset 750 mm, travel 1,060.66 mm**
   √(450² + 600²) = 750, 750 ÷ sin 45°
7. **8,932 mm**
   750 ÷ tan 4.8° = 8,931.51
8. **180 mm**
   18,000 × 1 ÷ 100
9. **Rise 1,767.25 mm, floor length 8,314.25 mm**
   8,500 × sin 12°, 8,500 × cos 12°
10. **14.43 mm, a 60° groove**
    25 × tan 30°
11. **Corner 144°, cut 72°, away from square 18°**
12. **Rise 1,169.71 mm, chord 3,785.26 mm**
    3,600 × tan 18°, 3,600 ÷ cos 18°
13. **29.74°**
    $\tan^{-1}(8 \div 14)$
14. **530.25 mm, 0.08 mm short**

Exercise 13C

1. 845.05 mm
2. 4.18 m
3. $12\frac{9}{16}$″ (12.586″)
4. 25.02°
5. 38.53° and 51.47°
6. $22\frac{5}{8}$″ (22.627″)
7. 2.39°
8. 9.21 mm
9. True offset 403.11 mm, travel 465.47 mm
10. 848.53 mm. Yes, it doubled: the set and the travel keep the same ratio.